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Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

SENIOR CERTIFICATE EXAMINATIONS/ NATIONAL SENIOR CERTIFICATE EXAMINATIONS

AGRICULTURAL SCIENCES P2

MAY/JUNE 2024

MARKING GUIDELINES

MARKS: 150

These marking guidelines consist of 10 pages.

SECTION A**QUESTION 1**

1.1	1.1.1	C ✓✓		
	1.1.2	B ✓✓		
	1.1.3	D ✓✓		
	1.1.4	D ✓✓		
	1.1.5	A ✓✓		
	1.1.6	B ✓✓		
	1.1.7	B ✓✓		
	1.1.8	C ✓✓		
	1.1.9	D ✓✓		
	1.1.10	A ✓✓	(10 x 2)	(20)
1.2	1.2.1	D ✓✓		
	1.2.2	H ✓✓		
	1.2.3	E ✓✓		
	1.2.4	J ✓✓		
	1.2.5	B ✓✓	(5 x 2)	(10)
1.3	1.3.1	Price elasticity of demand ✓✓		
	1.3.2	Income statement ✓✓		
	1.3.3	Heterozygote/crossbreed/hybrid ✓✓		
	1.3.4	Quantitative ✓✓		
	1.3.5	Prepotency ✓✓	(5 x 2)	(10)
1.4	1.4.1	Standardisation/grading ✓		
	1.4.2	Variable ✓		
	1.4.3	Epistasis ✓		
	1.4.4	Physical ✓		
	1.4.5	Inbreeding depression/degeneration ✓	(5 x 1)	(5)

TOTAL SECTION A: 45

SECTION B**QUESTION 2: AGRICULTURAL MANAGEMENT AND MARKETING****2.1 Marketing functions****2.1.1 Identification of the marketing function**

- A** Transportation ✓ (1)
- B** Storage ✓ (1)
- C** Packaging ✓ (1)

2.1.2 Guidelines for packaging

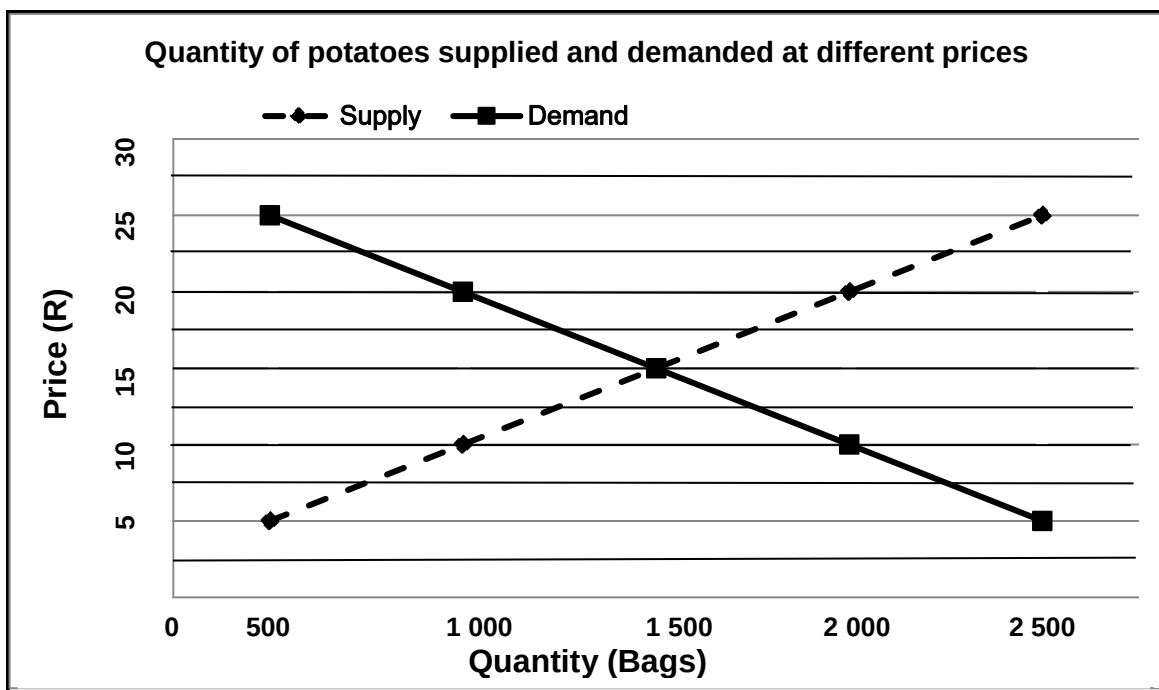
- E** Product identification ✓ (1)
- F** Containment ✓ (1)

2.1.3 TWO advantages of processing

- Increases the value of the product ✓
- A way of overcoming over-supply of product ✓
- Allows easier packaging, handling and transportation ✓
- Provides job opportunities ✓
- Enables the farmer to increase their share of final price paid for a product ✓
- Results in a product that has a longer shelf life ✓
- Reduces wastage of excess produce/prevent food spoilage ✓
- Improves food safety ✓
- Product is available throughout the year ✓ (Any 2) (2)

2.2 Supply, demand and price

2.2.1 Line graph



CRITERIA/RUBRIC/MARKING GUIDELINES

- Correct heading ✓
- X-axis: Correctly calibrated and labelled (Quantity) ✓
- Y-axis: Correctly calibrated and labelled (Price) ✓
- Correct units (Bags and R) ✓
- Line graph ✓
- Accuracy (80% + correct plotting) ✓ (6)

2.2.2 Law of supply

- The higher the price the more the quantities of potatoes supplied ✓✓
- The lower the price the lesser the quantities of potatoes supplied ✓✓ (Any 1) (2)

2.2.3 Condition in a market when the price is R15

- Market equilibrium/equilibrium point ✓ (1)

- 2.2.4 **TWO factors influencing the demand of potatoes other than price**
- Quality ✓
 - Consumer preferences/taste ✓
 - Income/buying power of consumers ✓
 - Number of consumers ✓
 - Price of competing/complementary/substitutes products ✓
 - Range of products available ✓
 - Use of the product ✓
 - Fashion ✓
 - Festive seasons ✓
 - Advertisement ✓
 - Legislation ✓
 - Sociological ✓
 - Research ✓
- (Any 2) (2)
- 2.3 **Approaches to agricultural marketing**
- A** Multi-segment ✓ (1)
- B** Green marketing ✓ (1)
- C** Niche marketing ✓ (1)
- 2.4 **Marketing channels of free marketing system**
- 2.4.1 **Identification of the marketing channel**
- Internet/online marketing channel ✓ (1)
- 2.4.2 **TWO advantages of the internet marketing channel for the farmer**
- A large market can be reached/global reach ✓
 - Money can be transferred before goods are delivered ✓
 - Faster method to reach the consumers ✓
 - Time saving/convenience ✓
 - Cost effective/less capital intensive ✓
 - Flexible/accessible anytime ✓
- (Any 2) (2)
- 2.4.3 **TWO marketing channels other than internet marketing**
- Farm gate ✓
 - Fresh produce ✓
 - Stock sales/auctions ✓
 - Direct/contact ✓
- (Any 2) (2)
- 2.5 **Factors hampering marketing of agricultural products**
- 2.5.1 Bulkiness ✓ (1)
- 2.5.2 Costs for intermediaries/middlemen ✓ (1)
- 2.5.3 Perishability/spoilage ✓ (1)
- 2.5.4 Risk/delays/spoilage/theft ✓ (1)
- 2.5.5 Seasonal fluctuations ✓ (1)

2.6 Entrepreneurial success factors

- | | | |
|-------|-------|-------------|
| 2.6.1 | B ✓ | (1) |
| 2.6.2 | D ✓ | (1) |
| 2.6.3 | C/D ✓ | (1) |
| 2.6.4 | A ✓ | (1) |
| | | [35] |

QUESTION 3: PRODUCTION FACTORS**3.1 Land****3.1.1 Letter representing the functions of land**

- | | | |
|-----|---|-----|
| (a) | Provision of food - C ✓ | (1) |
| (b) | Provision of space for infrastructure - B ✓ | (1) |

3.1.2 Economic characteristic of land

- | | | |
|---|-----------------------------|-----|
| A | Land is limited ✓ | (1) |
| D | Value of land appreciates ✓ | (1) |

3.1.3 TWO ways in which land productivity can be improved

- | | | |
|---|--|-------------|
| • | Water management/provision/irrigation ✓ | |
| • | Consolidating small uneconomical land units ✓ | |
| • | Farming land more efficiently/precision farming/
use scientific farming methods ✓ | |
| • | Improving soil fertility ✓ | |
| • | Changing cropping practices ✓ | |
| • | Restoring land potential ✓ | |
| • | Responsible use of pesticides ✓ | (Any 2) (2) |

3.1.4 The law of diminishing return

With the continued addition of more of one input to a productive process while other inputs are kept constant, a point will be reached where the output per unit of the added input will decline ✓✓ (2)

3.2 Labour productivity

- | | | |
|-------|--------------------------|-----|
| 3.2.1 | Motivation ✓ | (1) |
| 3.2.2 | Education ✓ | (1) |
| 3.2.3 | Recognition/motivation ✓ | (1) |
| 3.2.4 | Communication ✓ | (1) |

3.3 Labour skills**3.3.1 Indication of the best candidate as****(a) Person to manage the business optimally** - Candidate 2 ✓ (1)**(b) A technical advisor** - Candidate 1 ✓ (1)**3.3.2 TWO reasons for the answer in QUESTION 3.3.1 (a)**

- Higher scores/82% management skills/ability in management ✓
- Higher scores/80% interpersonal skills/ability in interpersonal skills ✓ (2)

3.3.3 Legislation regulating**(a) Safety**
Occupational Health and Safety Act/OHSA (No. 85 of 1993) ✓ (1)**(b) Participation in decision making**
Labour Relations Act/LRA (No. 66 of 1995) ✓ (1)**3.4 Capital****3.4.1 Types of capital****(a) C** ✓ (1)**(b) B** ✓ (1)**(c) A** ✓ (1)**3.4.2 ONE source of capital**

Financial institution/commercial banks/trust companies/Land bank/development bank/agricultural cooperatives/business partners/credit/production/grants/inheritance/family and friends/savings ✓ (1)

3.5 Balance sheet for the 28th of February 2023**3.5.1 Identification of the financial statement**

Balance sheet ✓ (1)

3.5.2 Provide the missing information for**(a) Value of the farm** ✓ (1)**(b) Bank overdraft** ✓ (1)**(c) R4 000 000** ✓ (1)**(d) R2 210 000** ✓ (1)

3.6 Income and expenditure graph**3.6.1 Identification of a farmer with highest income**

Farmer B ✓

(1)

3.6.2 Calculating the profit or loss for farmer A

Profit or loss = Total value of income – Total value of expenditure ✓

= R30 000 – R50 000 ✓

= – R20 000/loss ✓

(3)

3.7 Management**3.7.1 Management principles****(a)** Control ✓

(1)

(b) Implementation ✓

(1)

3.7.2 TWO external forces that affect a farming business

- Technological forces ✓
- Socio-cultural forces ✓
- Legal forces ✓
- Political forces ✓
- Ethical forces ✓
- Economic forces ✓
- Environmental forces ✓
- Competitive forces ✓

(Any 2)

(2)

[35]**QUESTION 4: BASIC AGRICULTURAL GENETICS****4.1 Monohybrid crossing****4.1.1 Punnett square**

Gametes	B	b
b	Bb	bb
b	Bb	bb

CRITERIA FOR MARKING

- Gametes of parent 1 ✓
- Gametes of parent 2 ✓
- Genotype of offspring ✓
- Punnet square populated with gametes and offspring genotypes ✓

(4)

4.1.2 Calculation of the percentage of pink flowers

- = $2 \div 4 \times 100$ ✓
- = 50% ✓

(2)

4.1.3 Calculate the number of heterozygous offspring

- $= 2 \div 4 \times 350$ ✓
- $= 175$ ✓

(2)

4.2 Monohybrid crossing presentation**4.2.1 Pattern of inheritance shown by Pair 1**

Incomplete dominance/co-dominance ✓

(1)

4.2.2 Justification for co-dominance/incomplete for pair 1

The alleles for both parents are dominant ✓

(1)

4.2.3 The phenotype of Pair 2 offspring

All round fruits ✓

(1)

4.2.4 Genotypic ratio of Pair 2 offspring

1AA : 1Aa ✓

(1)

4.3 Crossing schematic representation**4.3.1 Mendel law that applies in the crossing**

The law of segregation ✓

(1)

4.3.2 The process that resulted to (7)

Fertilization ✓

(1)

4.3.3 Genotypes and phenotypes**(a) (2) White** ✓

(1)

(7) Dark ✓

(1)

(b) (1) DD ✓

(1)

(6) d ✓

(1)

4.4 Variation**4.4.1 Genetic term**

Variation ✓

(1)

4.4.2 TWO other internal factors

- Mutation ✓
- Random fertilization ✓

(2)

4.4.3 TWO methods of selecting farm animals

- Progeny selection ✓
- Family selection ✓
- Pedigree selection ✓
- Mass selection ✓

(Any 2)

(2)

4.5 Crossbreeding Holstein cows with Dairy Swiss bulls**4.5.1 The breeding system**

Crossbreeding ✓ (1)

4.5.2 Reason for crossbreeding

Two different/unrelated breeds were used ✓ (1)

4.5.3 TWO advantages of crossbreeding from the case study

- An improvement in the growth rate of the calves ✓
- Offspring were better adapted to hot conditions ✓
- Udders and legs improved ✓ (Any 2) (2)

4.5.4 Differentiation**Inbreeding**

Mating of animals that are related to one another ✓ (1)

Line breedingMating of animals that are closely related to an outstanding ancestor/
bull ✓ (1)**4.5.5 ONE example for each of the breeding systems****Inbreeding** - Father and daughter/son and mother/brother and sister ✓ (1)**Line breeding** - A bull with heifers from the second generation ✓ (1)**4.6 Genetically modified organisms****4.6.1 TWO potential risks of genetically modified crops to the environment**

- Can produce super weeds ✓
- Insect resistant plants can kill beneficial insects ✓
- Use of excessive amounts of herbicides may damage the soil ✓
- Loss of biodiversity/useful plants can be destroyed ✓
- Reduce the effectiveness of herbicides/pesticides ✓ (Any 2) (2)

4.6.2 TWO benefits of genetically modified crops

- Environmental benefits ✓
- Health benefits ✓
- Economic benefits ✓ (Any 2) (2)

[35]

TOTAL SECTION B: 105
GRAND TOTAL: 150